UNIVERSITY OF PLYMOUTH

PUSL3190 Computing Project

Project Proposal

"TruthLens"- Fake News Detection and News Summarization Mobile Application

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1 Chapter 01 – Problem Statement

1.1 Introduction

With the rapid growth of social media, online journalism, and digital content platforms, the spread of fake news has emerged as a critical global issue. False or misleading information can travel faster than verified content, influencing public perception, political outcomes, and social behaviors. Many users encounter manipulated headlines or fabricated stories yet lack the tools to quickly verify authenticity.

Traditional fact-checking methods rely on manual review by experts, which can be time-consuming and cannot scale to the speed at which news spreads online. Meanwhile, users consume and share information instantly, accelerating the dissemination of false content. Many existing solutions present results in complex, technical language, making them inaccessible to general users. (Ahmed, 2020)

In Sri Lanka, where internet penetration and social media usage are high, the lack of localized fake news detection tools has heightened the problem. People frequently rely on unverified sources, leading to misinformation-driven behaviors and public confusion. This situation is compounded by the absence of integrated solutions that combine news verification with summarization for ease of consumption.

TruthLens addresses these issues by providing a mobile application that detects fake news using advanced AI and NLP techniques while also summarizing content for users. By leveraging Flutter for cross-platform mobile development, Laravel for backend API management, and Firebase for real-time database and authentication, the system ensures scalability, security, and usability.

2 Chapter 02 – Project Description

2.1 Overview

TruthLens is an AI-driven mobile application designed to detect fake news and provide concise summaries for easier understanding (Zhang, 2022). Users can input news text or links, and the application evaluates credibility while presenting an automated summary.

The system uses natural language processing (NLP) and deep learning to identify patterns, sentiment, and source reliability. By combining fake news detection and summarization in a single platform, TruthLens addresses both misinformation and information overload. The mobile-first approach ensures accessibility and engagement, while the backend stack (Laravel + Firebase) supports efficient data handling, authentication, and storage.

2.1.1 Stakeholders

- General Users: Individuals seeking to verify news authenticity quickly.
- Students and Researchers: Users needing summarized and verified news for academic purposes.
- Journalists and Media Analysts: Professionals filtering reports efficiently.
- Developers and AI Researchers: Interested in advancing AI Based verification tools.
- Academic Supervisors: Technical Development and ethical compliance.

2.2 Project Objectives

1. Develop a mobile application capable of detecting fake news using NLP and deep learning techniques.
2. To implement an automated news summarization module for generating concise and factual summaries.
3. To visualize credibility scores and classification results in a simple, intuitive interface.
4. To enhance user trust and understanding through transparency and explanation-based AI outputs.
5. To create a scalable, ethical, and user-centered mobile solution adaptable to multiple languages and platforms.

2.3 Objective Explanations

1. Accuracy:
   The fake news detection model will be trained on authentic and fabricated news datasets to achieve high accuracy in classifying content as real or fake. Continuous fine-tuning will improve performance across diverse topics.
2. Efficiency:
   The news summarization component will condense lengthy articles into key points, enabling users to save time and consume verified news efficiently.
3. Usability:
   The mobile app will provide a clean, accessible interface with real-time feedback. Results will be displayed using clear visual indicators and color-coded credibility scores.
4. Scalability:
   Cloud-based architecture using Firebase ensures scalability for a large number of users while maintaining system performance.
5. Ethical Transparency:
   TruthLens emphasizes transparency by explaining the reasoning behind each credibility assessment, ensuring users understand how conclusions are reached.

Keywords: Fake News Detection, Natural Language Processing, Deep Learning, News Summarization, Artificial Intelligence.

3 Chapter 03 – Research Gap

3.1 Literature Review & Background

Many existing tools, such as PolitiFact, Snopes, and FactCheck.org, rely heavily on manual verification by human experts, limiting real-time usability (Ahmed H. , 2020). Although these platforms provide highly accurate fact-checking, the delay between publication and verification limits their real-time usability. Recent studies on AI-based fake news detection have shown promising outcomes; however, most focus primarily on binary text classification (true/false) and neglect crucial aspects such as explainability, contextual reasoning, and user trust.

Additionally, most of the current solutions lack integration between fake news detection and news summarization (Zhang, 2022). Users are often required to switch between different tools to verify and summarize content, leading to inefficiency and reduced engagement. While some research prototypes combine machine learning with natural language processing (NLP) for misinformation detection, few systems provide interpretable feedback that helps users understand why an article is flagged as fake or misleading.

Another critical gap is the language and contextual limitation of most existing systems. Many are designed exclusively for English-speaking audiences and do not adapt well to regional linguistic nuances, local misinformation patterns, or mobile-first usage. Furthermore, datasets used in fake news detection research, such as LIAR or FakenewsNet, are predominantly Western-centric, which limits the generalization of models to diverse cultural and informational contexts.

The TruthLens project aims to bridge these gaps by integrating real-time fake news detection, abstractive summarization, and explainable reasoning into a single, mobile-accessible platform. The system leverages transformer-based architectures for both classification and summarization while ensuring transparency through confidence scores and evidence highlighting. This holistic approach not only enhances accuracy but also improves user trust, reduces verification time, and broadens accessibility across different languages and user demographics.

3.2 Comparison of Existing Systems vs Proposed System

Table 3.1: Comparison of Existing Systems vs Proposed System

Feature	PolitiFact	Snopes	FactCheck.org	TruthLens
AI Based Detection	✕	✕	✓	✓
Automated Summarization	✕	✕	✕	✓
Real Time Analysis	✕	✕	✕	✓
Mobile Accessibility	✓	✓	✕	✓
Explanation Transparency	✕	✕	✕	✓
Multi Language Support	✕	✕	✕	✓

TruthLens integrates detection and summarization in a single, accessible system, offering features absent in current global solutions.

3.3 Methodology

The project will adopt the Agile (Scrum) methodology, allowing iterative development, testing, and feedback integration.

Sprint 1: Requirement gathering, literature review, and dataset collection.
Sprint 2: Design and development of the fake news detection model.
Sprint 3: Integration of summarization algorithms and mobile frontend.
Sprint 4: Testing, refinement, and user evaluation.

Continuous collaboration with supervisors and peer reviewers will ensure that the system remains accurate, efficient, and user-friendly.

3.4 Algorithms and Techniques

- Data Collection:
  The system will utilize publicly available benchmark datasets such as Kaggle's FakeNewsNet and the LIAR dataset to ensure diversity and credibility in training and validation (Ahmed H. , 2020). These datasets include labeled news articles and statements that are essential for supervised learning and fake news detection.
- Preprocessing:
  Collected textual data will undergo preprocessing steps including tokenization, lemmatization, stop-word removal, and feature extraction. This ensures that the data fed into the models are clean, normalized, and semantically consistent, enabling more accurate model training and inference.
- Modeling:
  Deep learning architectures such as BERT, LSTM, and RoBERTa will be employed for text classification tasks. These models will be implemented and fine-tuned using the Hugging Face Transformers library, which provides pre-trained state-of-the-art natural language processing (NLP) models. The use of Hugging Face allows efficient integration of transfer learning techniques, accelerating development while maintaining high performance in text understanding and classification.
- Summarization:
  For generating concise and meaningful summaries of detected articles, transformer based models such as T5 and BART will be utilized through the Hugging Face framework. These models are capable of abstractive summarization (Zhang, 2022), producing human-like summaries that enhance content comprehension for users.
- Integration:
  The backend will be implemented using Firebase for data storage, authentication, and cloud processing, while the Laravel API layer will manage communication between the Flutter frontend and the AI models hosted on Hugging Face. This integration ensures seamless mobile interactions, efficient real-time updates, and secure user data handling.
- Evaluation Metrics:
  Model performance will be evaluated using multiple metrics including Accuracy, Precision, Recall, F1-score, Latency, and User Feedback. These metrics will provide both quantitative and qualitative insights into the system's effectiveness and responsiveness.

4 Chapter 04 – Requirement Analysis

4.1 Functional Requirements

- The app shall allow users to paste or input news content or URLs.
- The system shall classify news as real or fake using NLP-based models.
- The app shall generate automatic summaries of the given news article.
- The app shall display credibility scores, reasoning, and sentiment analysis.
- The app shall maintain a history of analyzed news for user reference.

4.2 Non-Functional Requirements

- The system should respond within 5 seconds for average news length.
- The interface should be mobile-responsive and easy to navigate.
- All user data shall be stored securely using Firebase authentication.
- The system should achieve at least 90% model accuracy (Ahmed, 2020).
- The app should run efficiently on Android devices with moderate specifications.

4.3 User Requirements

- Users should verify news authenticity through a simple interface.
- Users should access concise summaries for better comprehension.
- Users should understand credibility scores and reasons behind detection.
- The app should support multi-language compatibility in the future.

4.4 Hardware and Software Requirements

Table 4.1: Hardware & Software Requirements

Hardware Requirements	• Android Smartphone (Minimum 3GB RAM, 32GB storage) for testing and deployment. • Laptop or PC with at least 8GB RAM, multi-core processor, and stable internet connection for development and model training.
Software Requirements	• Programming Languages: Dart (for Flutter), PHP (for Laravel), and Python (for AI model development). • Frameworks & Libraries: Flutter SDK, Laravel Framework, TensorFlow / PyTorch for machine learning, and Hugging Face Transformers for NLP model integration. • Database: Firebase • APIs: Hugging Face Inference API, News Verification APIs, and Firebase Authentication APIs. • Development Tools: VS Code • Version Control: GitHub • Design Tool: Figma • Data Sources & Platforms: Kaggle (for datasets)

4.5 Methods to Collect Requirements

The requirements should be derived from real user needs. To ensure that the following methods will be used.

1. User Interviews: Collect insights from students and journalists about fake news issues.
2. Surveys: Gather user expectations for interface design and summarization clarity.
3. Competitor Review: Evaluate current fact-checking and summarization tools to identify usability gaps.

5 Chapter 05 – External Organizations

The project is primarily carried out independently under the supervision of the university's academic staff. No direct external funding or formal industrial partnership is currently involved. However, the development and evaluation phases leverage reputable online datasets and open-source frameworks contributed by global research communities to enhance reliability and innovation.

Specifically, platforms such as Kaggle, Hugging Face, and Google Dataset Search provide high-quality, publicly available datasets related to fake news detection, news classification, and text summarization. These resources will be used strictly for academic and non-commercial purposes, adhering to each dataset's respective licensing and ethical-use policies.

Additionally, open-source libraries like TensorFlow, PyTorch, and Transformers will be integrated for machine learning model implementation. These frameworks are maintained by international organizations and research groups that foster collaborative development and transparency, aligning with the project's goal of reproducibility and ethical AI practice.

Overall, while TruthLens remains an independent academic project, it acknowledges and builds upon the collective knowledge and contributions of the wider open source and research community, ensuring a transparent, ethical, and academically sound development process.

6 Chapter 06 – Project Timeline

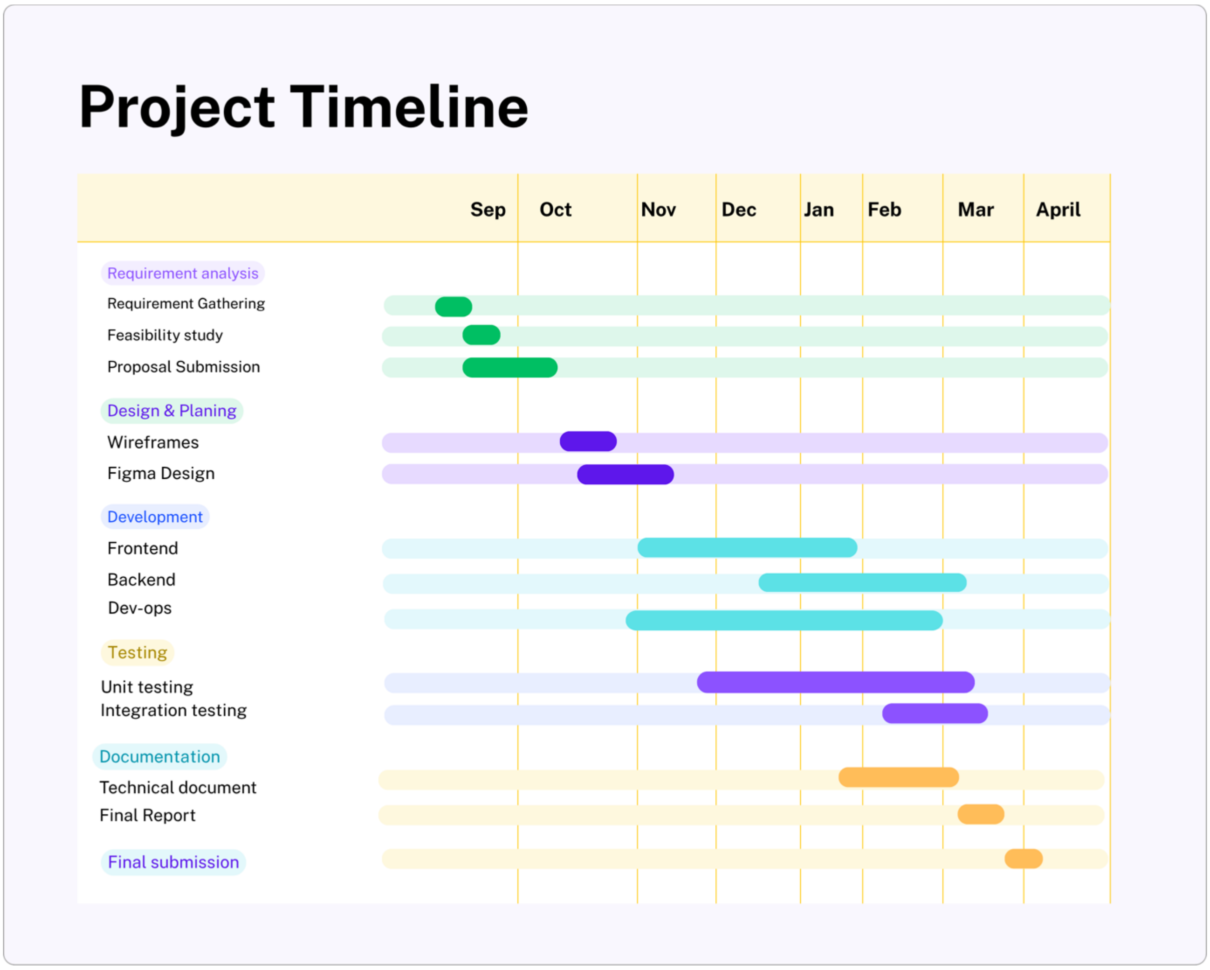

Figure 6.1: Project Timeline

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